



EDUCATION

M.Sc. (Hons.) Mathematics / 4 Years

Birla Institute of Technology and Science, Pilani - K. K. Birla Goa Campus

M.A. Linguistic - Text Mining / 1 Year

Vrije Universiteit Amsterdam

WORK EXPERIENCE

Thesis / Ongoing / Textkernal & VU AMSTERDAM

RQ - How can university curricula and course descriptions be leveraged to enrich a knowledge graph of profession-skill-degree relationships?

Data Scientist/ 2.5 Years / Ralph Lauren

Build and deployed a model exposed through an API endpoint in Dataiku that managed Data scraping and auto-filling a form for a supply chain system. The model used OCR Tesseract capabilities along with custom Natural Language Understanding and processing to extract the required dates, country, name of certifying entity, type of audit form and certificate number along with which product the form was applicable to. It was a dynamic code that could handle multiple formats and form types. It also had a built-in error handling system and email notification capacity. The model was deployed with an accuracy of 97%.

Automated image extraction and labelling from pdfs and excels for the product team helped reduce 1 week's work for a 5 person team to a couple of minutes. The python code was also later used by other teams for similar work. Created the predictive code (python) that helped estimate the cost of raw materials, packing, shipping and cost to the company to produce various products. The POC was picked up within two weeks, built and put into use within one build cycle. Used various techniques like feature engineering, data cleaning and augmentation and clustering as well as hyper-tuning the model to increase the accuracy of the code by 20% (70% to 90%). The same approach and large parts of the code were reusable for other models.

Developed and managed multiple reports for various teams in MicroStrategy. Completed 13 user stories for such reports with a 92% completion rate. Created new data fields for the system - created the logic, helped update and test the backend code for the calculation of the new fields as well as the code for the data pipeline(AWS) from the data lake(AWS redshift) into MSTR with a test case coverage of 97%.

Assigned to a third-party team integration team(for a custom demand management system) in SSIS for 2 months at 50% capacity. I was able to build 3 different integrations with 2 different systems. I was previously also a part of the team that planned and designed the integration architecture and specs for the same as well as cleaned and managed the data flow during the initial go-live for the same custom system. Worked in the team importing and exporting/managing data into a new cloud-based system for the vendor management system while shifting technologies. Wrote data cleaning and wrangling code in R (for data transfer to the new system) to remove the manual error and reduce manpower by 80%.

Management Trainee/ 4 Months / Cogoport. Pvt. Ltd.

Devised and executed the pan India email campaign strategy based on data analysis and segmented targeting, resulting in 20% increase in open rate of emails. Researched and analyzed monthly commodity reports to come up with strategies to minimize logistics costs for commodity dealers.

TECHNICAL PROFICIENCY

Python / Tableau / R programming / Microsoft Excel / SQL / MicroStrategy / SSIS / Agile / Jira / TestRail / Mural

PROJECTS.....

Semantic Role Labelling / 2 Months / Study Project for VU AMSTERDAM

Feature engineering for testing a simple ML model for Semantic role labelling. Comparative study and exploration of BERT SRL fine-tuning.

Named Entity Recognition / 2 Months / Study Project for VU AMSTERDAM

Feature engineering for an end-to-end comparative study of various ML models for NER. Comparative research and exploration of BERT and BiLSTM for NER.

Language as Data / 2 Months / Study Project for VU AMSTERDAM

Study of the annotation process for NLP and biases found. Scrapping data and annotating it manually. Study of licensing and ethical issues in NLP. Designed and pitched the full pipeline and features for a model that graphs the coherence of dementia patients' spontaneous speech.

Negation Cue Detection / 1 Month / Study Project for VU AMSTERDAM

Feature engineering for an Negation Cue Detection as part of a 4 person group working in an agile methodology.

User Profiling / 4 Months / Cybage Software Pvt. Ltd.

Worked on a Microsoft project that analysed users' personal data to segment user personas for targeted customer service calls with the end goal of increasing the company's overall NPS. Cluster analysis and feature importance of various user-centric databases were studied.

Topic Modelling / 4 Months / Cybage Software Pvt. Ltd.

Sorted through customer feedback from a huge data set of Microsoft text reviews to tag feed backs by their topic e.g pricing, software, support, etc. Cleaned data with techniques like stemming, removing filler words/numbers, etc. Used an unsupervised algorithm to get a cluster of 15 main topics to find the most common reason for complaints. Successfully implemented the above to get an coherence score of 83%.

Sentiment Analysis / 2 Months / Cybage Software Pvt. Ltd.

Sorting negative customer feedback from a huge data set of Microsoft text reviews to tag feed backs as net positive or negative. Cleaned data with techniques like stemming, removing filler words/numbers, etc. Used a lexicon based approach mixed with a weighted n-gram to get the overall sentiment of each review to circumvent the errors we got due to double negatives/ metaphors etc. Successfully implemented the above to get an accuracy of 85%.

Data Exploration and Visulalization / 2 Months / Cybage Software Pvt. Ltd.

Predictive Modelling / 2 Months / Anukoolan

Data Scraping and Cleaning / 2 Months / Dealroom.co

ELECTIVES/ CERTIFICATIONS.....

Cryptography / Game theory / Graphs and Networks / Machine Learning / Numerical Analysis / Operations Research / Optimization / Principles of Management / Statistical Inference / Probability/ Graph Theory / Python - **BITS GOA**

Transformers / Large Language Models / Data Scrapping / Data Annotation / FrameNet / Verbnet / WordNet / Interpretability / Data Cleaning / MySQL - **VU AMSTERDAM**

Data Scientist career track with R programming - **DATA CAMP**